

CONTACT	Center for Gravitational Physics, Department of Physics, University of Texas at Austin, Austin, TX, 78712 USA	Personal website jay.wadekar@utexas.edu
PROFESSIONAL POSITIONS	Assistant professor <i>The University of Texas at Austin, Department of Physics, Austin, TX</i> Jan 2026 - current Affiliated faculty <i>Oden Institute for Computational Engineering and Sciences, UT Austin</i> 2026 - current Postdoctoral fellow <i>Johns Hopkins University (JHU), Baltimore, MD</i> Sep 2024 - Dec 2025 Member (postdoctoral fellow) <i>Institute for Advanced Study (IAS), Princeton, NJ</i> Sep 2021 - Aug 2024	
EDUCATION	New York University (NYU), New York, NY Ph.D. (alongside MS & M.Phil in Astrophysics) September 2021 Indian Institute of Technology, Bombay (IITB), Mumbai, India B.Tech (Bachelor of Technology) in Engineering Physics with Honors in Physics May 2015	
RESEARCH INTERESTS	- AI/ML: agentic LLMs (scientific discovery, validation), physics-informed ML, interpretability - Compact objects: gravitational wave searches and progenitor inference - Cosmology with Sunyaev-Zeldovich (SZ) and galaxy spectroscopic surveys - Dark matter phenomenology from observations of dwarf galaxies	
AWARDS & HONORS	• Membership, IAS, Princeton (2021 - 2024) • Weinberg Institute Postdoctoral Fellowship, UT Austin (declined) • Subrahmanyan Chandrasekhar postdoctoral fellowship, Perimeter Institute (declined) • James Arthur Dissertation Fellowship, NYU (2020 - 2021) - awarded to one student across all science, humanities, social science programs at NYU. • James Arthur Graduate Fellowship, NYU (2019 - 2020) • Henry Mitchell McCracken Fellowship at NYU (2015 - 2019) • All India Rank 139 in IIT-JEE 2011 exam (99.97 percentile) among 485,000 candidates. • KVPY fellowship (Kishore Vaigynaik Protsahan Yojana) by the Govt. of India (declined) • NTSE fellowship (National Talent Search Scholarship) by the Govt. of India. • Travel grants: DAP travel award (600\$) & DGRAV travel award (300\$) for APS April Meeting 2019. DAP travel award (600\$) for APS April Meeting 2018	
INVITED TALKS & COLLOQUIA	Gravity seminar, UIB, Spain (remote) November 2025 Astrophysics seminar, University of Pennsylvania September 2025 Foundational AI workshop, Vanderbilt University August 2025 Physics seminar, IIT Bombay June 2025 Astrophysics seminar, TIFR, India June 2025 Physics seminar, UT Austin February 2025 DGRAV-APS seminar (remote) [video] October 2024 Gravity, Astro and Particle Physics seminar, Penn state University February 2024 CTC seminar, University of Maryland February 2024 Astrophysics seminar, New York University January 2024 BHI foundations seminar, Harvard [video] November 2023 Lunch seminar, Carnegie observatories November 2023 N3AS seminar, UC Berkeley (remote) [slides] September 2023 Theoretical physics seminar, TIFR, India September 2023 Astrophysics seminar, ICTS, India August 2023 Astrophysics seminar, IAS, Princeton February 2022	

Astrophysics seminar, IIT Hyderabad, India [slides]	February 2022
Astrophysics seminar, TIFR, India	January 2022
SOTU seminar, TIFR, India	November 2021
RPM seminar, Lawrence Berkeley National Lab, CA [slides]	January 2021
CCA lunch talk, Center for computational astrophysics, NY	August 2020
Princeton/IAS Cosmology lunch talk , Princeton, NJ	December 2019
Cosmology seminar, TIFR, Mumbai, India	December 2019
Cosmology seminar, UC Berkeley, CA [slides]	October 2019
Workshop on dynamics of LSS formation, MIAPP, Garching, Germany	July 2019

MENTORING

- Mark Cheung: JHU graduate student. *Co-authored three papers.*
- Zihui Wang: NYU graduate student. *Co-authored three papers.*
- Ana Maria Delgado: Harvard graduate student. *Co-authored a paper.*
- Leander Thiele: Princeton graduate student. *Co-authored three papers.*
- Zihan Zhou: Princeton graduate student. *One paper in prep.*
- Abby Mintz: Princeton graduate student. *One paper in prep.*
- Konstantinos Kritos: JHU graduate student. *One paper in prep.*
- Sophia Yi: JHU graduate student. *Co-authored one paper.*
- Arush Pimpalkar: NIT Trichy undergraduate student. *Co-authored one paper.*
- Tibor Rothschild: Yale undergraduate student.
- Param Gogia: Vassar college undergraduate student.

PUBLICATIONS

The most-updated list and metrics are available in the [ADS library](#).
I have published over 45 papers, 1600+ citations, h-index 20
32 first/second-author papers, 840 citations ([library link](#))

FIRST AUTHOR

- Improving gravitational wave search sensitivity with TIER: [arXiv:2507.08318](#)
Trigger Inference using Extended strain Representation
D. Wadekar, A. Pimpalkar, M. H. Y. Cheung, B. Wandelt, E. Berti, et al.
- New search pipeline for gravitational waves with higher-order modes [arXiv:2405.17400](#)
using mode-by-mode filtering
PRD 2024
D. Wadekar, J. Roulet, T. Venumadhav, A. Mehta, B. Zackay, et al.
- New black hole mergers in the LIGO-Virgo O3 data from [arXiv:2312.06631](#)
a gravitational wave search including higher-order harmonics
under review
D. Wadekar, J. Roulet, T. Venumadhav, A. Mehta, B. Zackay, et al.
- New approach to template banks of gravitational waves with higher harmonics: [arXiv:2310.15233](#)
reducing matched-filtering cost by over an order of magnitude
PRD 2024
D. Wadekar, T. Venumadhav, A. Mehta, J. Roulet, et al.
- The SZ flux-mass ($Y - M$) relation at low halo masses: improvements with [arXiv:2209.02075](#)
symbolic regression and strong constraints on baryonic feedback
MNRAS 2023
D. Wadekar, L. Thiele, F. Villaescusa-Navarro, J. C. Hill, D. Spergel, et al.
- Augmenting astrophysical scaling relations with machine learning: [arXiv:2201.01305](#)
application to reducing the SZ flux-mass scatter
PNAS 2023
D. Wadekar, L. Thiele, F. Villaescusa-Navarro, J. C. Hill, D. Spergel, et al.
- Strong constraints on decay and annihilation of dark matter [arXiv:2111.08025](#)
from heating of gas-rich dwarf galaxies
PRD 2022
D. Wadekar*, Z. Wang*
- Modeling the neutral hydrogen assembly bias with machine learning [arXiv:2012.00111](#)
and symbolic regression
D. Wadekar, F. Villaescusa-Navarro, S. Ho, L. Perreault-Levasseur
- Cosmological constraints from BOSS with analytic covariance matrices [arXiv:2009.00622](#)
PRD 2020
D. Wadekar, M. Ivanov, R. Scoccimarro
- HI-net: Generating neutral hydrogen from dark matter with neural networks [arXiv:2007.10340](#)
ApJ 2021
D. Wadekar, F. Villaescusa-Navarro, S. Ho, L. Perreault-Levasseur

3. Gas-rich dwarf galaxies as a new probe of dark matter interactions with ordinary matter [arXiv:1903.12190](#)
PRD 2021
D. Wadekar, G. Farrar
2. The Galaxy Power Spectrum Multipoles Covariance in Perturbation Theory [arXiv:1910.02914](#)
PRD 2020
D. Wadekar, R. Scoccimarro [Editors' suggestion]
1. Zeldovich pancakes at redshift zero: the equilibration state and phase space properties. MNRAS 2015
D. Wadekar, S. Hansen [[arXiv:1411.6627](#)]

* indicates alphabetical authorship

SECOND/
THIRD
AUTHOR

24. Unified remnant models for aligned-spin, precessing, and eccentric binary black hole mergers [arXiv:2608.00934](#)
T. Islam, D. Wadekar, G. Khanna
23. The diffraction-lensing interpretation of GW231123 with astrophysical priors [arXiv:2607.21834](#)
M. H. Y. Cheung, D. Wadekar, M. Zaldarriaga, T. Venumadhav, J. Roulet, et al.
22. Improving low-latency multi-messenger follow-up of neutron star-black hole mergers with mode-by-mode filtering [arXiv:2606.17137](#)
F. Iacovelli, D. Wadekar, J. Roulet, E. Berti, A. Corsi
21. Predicting intermediate-mass black hole formation in star clusters with machine learning [arXiv:2605.21593](#)
K. Kritos, D. Wadekar, E. Berti
20. Discovery of interpretable surrogates via agentic AI: application to gravitational waves [arXiv:2605.11280](#)
T. Islam, D. Wadekar, T. Venumadhav, M. Zaldarriaga, A. K. Mehta, et al.
19. gwBenchmarks: stress-testing LLM agents on high-precision gravitational wave astronomy [arXiv:2605.11269](#)
T. Islam, D. Wadekar, Z. Zhou
18. GW190711_030756 and GW200114_020818: astrophysical interpretation of two asymmetric binary black hole mergers in the IAS catalog [arXiv:2604.07388](#)
T. Islam, T. Venumadhav, D. Wadekar, A. K. Mehta, J. Roulet, et al.
17. Kick matters: the impact of a new recoil model on the retention of hierarchical black-hole remnants in globular clusters [arXiv:2603.10170](#)
T. Islam, D. Wadekar, K. Kritos
16. Searching for precessing binary systems with mode-by-mode filtering and marginalization [arXiv:2603.05784](#)
Z. Zhou, D. Wadekar, J. Roulet, O. Ivashtenko, T. Venumadhav, et al.
15. Progenitor of the recoiling supermassive black hole RBH-1 identified using HST and JWST imaging [arXiv:2601.18986](#)
PRL 2026
T. Islam, T. Venumadhav, D. Wadekar
14. Sample-efficient non-Gaussian noise reduction in gravitational wave data via learnable wavelets [arXiv:2601.14326](#)
PRD 2026
A. Pimpalkar, D. Wadekar, M. H. Y. Cheung, E. Berti
13. Accurate models for recoil velocity distribution in black hole mergers with comparable to extreme mass-ratios and their astrophysical implications [arXiv:2511.11536](#)
PRD 2026
T. Islam, D. Wadekar
12. Generating optimal gravitational-wave template banks with metric-preserving autoencoders [arXiv:2511.10466](#)
G. Cabass, D. Wadekar, M. Zaldarriaga, Z. Zhou
11. Binary black hole population inference combining confident and marginal events from the IAS-HM search pipeline [arXiv:2508.15350](#)
PRD 2025
A. K. Mehta, D. Wadekar, I. Anantpurkar, J. Roulet, T. Venumadhav, et al.

10. Searching for intermediate mass ratio binary black hole mergers in the third observing run of LIGO-Virgo-KAGRA arXiv:2507.01083
PRD 2026
M. H. Y. Cheung, D. Wadekar, A. K. Mehta, T. Islam, J. Roulet, et al.
9. Significant increase in sensitive volume of a gravitational wave search upon including higher harmonics arXiv:2501.17939
A. Mehta, D. Wadekar, J. Roulet, I. Anantpurkar, T. Venumadhav, et al.
8. New binary black hole mergers in the LIGO-Virgo O3b data arXiv:2311.06061
PRD 2025
A. Mehta, S. Olsen, D. Wadekar, J. Roulet, T. Venumadhav, et al.
7. Fast marginalization algorithm for optimizing gravitational wave detection, parameter estimation and sky localization arXiv:2404.02435
PRD 2024
A. Mehta, S. Olsen, D. Wadekar, J. Roulet, T. Venumadhav, et al.
6. In Pursuit of Love: First Templated Search for Compact Objects with Large Tidal Deformabilities in the LIGO-Virgo Data arXiv:2306.00050
PRD 2024
H. S. Chia, T. Edwards*, D. Wadekar*, A. Zimmerman*, et al.*
5. Constraining axion and compact dark matter with interstellar medium heating arXiv:2211.07668
PRD 2023
D. Wadekar, Z. Wang**
4. Percent-level constraints on baryonic feedback with CMB spectral distortions arXiv:2201.01663
PRD 2022
L. Thiele, D. Wadekar, J. C. Hill, N. Battaglia, J. Chluba, et al.
3. Modeling the galaxy-halo connection with machine learning arXiv:2111.02422
MNRAS 2022
A. Delgado, D. Wadekar, B. Hadzhiyska, S. Bose, L. Hernquist, S. Ho
2. Comment on “Calorimetric Dark Matter Detection with Galactic Center Gas Clouds”
*G. Farrar, F. Lockman, N. McClure-Griffiths, D. Wadekar** [arXiv:1903.12191] PRL 2020
1. Variance Adaptation in Navigational Decision Making eLife 2018
R. Gepner, J. Wolk, D. Wadekar, S. Dvali, M. Gershow

* indicates alphabetical authorship

Nth-AUTHOR

11. HI simulations for cosmology with the SKA Observatory arXiv:2606.25969
T. Ronconi et al. (incl. D. Wadekar)
10. Not too close! Evaluating the impact of the baseline on the localization of binary black holes by next-generation gravitational-wave detectors arXiv:2604.11871
F. Iacovelli, L. Reali, E. Berti, A. Corsi, B. Sathyaprakash, et al. (incl. D. Wadekar)
9. Systematic biases in parameter estimation on LISA binaries. II. The effect of excluding higher harmonics for spin-aligned, high-mass binaries arXiv:2602.09088
PRD 2026
S. Yi, F. Iacovelli, E. Berti, R. S. Chandramouli, S. Marsat, et al. (incl. D. Wadekar)
8. Data-driven extraction and phenomenology of eccentric harmonics in eccentric spinning binary black hole mergers arXiv:2509.20556
T. Islam, T. Venumadhav, A. K. Mehta, D. Wadekar, J. Roulet, et al.
7. Sampler-free gravitational wave inference using matrix multiplication arXiv:2507.16022
PRD 2025
J. Mushkin, J. Roulet, B. Zackay, T. Venumadhav, et al. (incl. D. Wadekar)
6. Data-driven extraction, phenomenology, and modeling of eccentric harmonics in binary black hole merger waveforms arXiv:2504.12469
PRD 2025
T. Islam, T. Venumadhav, A. K. Mehta, I. Anantpurkar, D. Wadekar, et al.
5. gwharmon: first data-driven surrogate for eccentric harmonics in binary black hole merger waveforms arXiv:2504.12420
T. Islam, T. Venumadhav, A. K. Mehta, I. Anantpurkar, D. Wadekar, et al.
4. Systematic biases from the exclusion of higher harmonics in parameter estimation on LISA binaries arXiv:2502.12237
S. Yi, F. Iacovelli, S. Marsat, D. Wadekar, E. Berti
3. The CAMELS project: public data release arXiv:2201.01300
F. Villaescusa-Navarro et al. (incl. D. Wadekar)

	2. The CAMELS Multifield Dataset: Learning the Universe's Fundamental Parameters with Artificial Intelligence <i>F. Villaescusa-Navarro et al. (incl. D. Wadekar)</i>	arXiv:2109.10915 ApJ 2022
	1. The CAMELS project: Cosmology and Astrophysics with Machine Learning Simulations <i>F. Villaescusa-Navarro et al. (incl. D. Wadekar)</i>	arXiv:2010.00619 ApJ 2021
PRESS	New Scientist , Phys.org , Cosmos magazine , Eurekalert , Science daily , Space Ref , Tech times , Scienmag , Tech explorerist , IAS , CCA , UConn	
SERVICE	- Co-chair of NASA Cosmic Origins AI/ML STIG - Organizer of the ML×Astronomy meetings at JHU and STScI - Organizer of the IAS astrophysics seminars - Organizer of the Dark Cosmos seminar series at Princeton University - Referee for MNRAS, Phys. Rev. D, Annalen der Physik. - Author of the public CovaPT code for calculating analytic covariance matrices for galaxy spectroscopic surveys. - Author of the public IAS-HM pipeline code for finding black hole mergers in public gravitational wave strain data.	Fall 2025-current Fall 2024-2025 Fall 2023-2024 Fall 2022-2023
TEACHING EXPERIENCE	- Teaching Assistant (TA) at NYU for Mathematical Physics (undergraduate) - TA at NYU for Electricity & Magnetism- I (undergraduate) - TA at IITB for Electromagnetism- I (undergraduate)	Spring 2018 Fall 2016 Spring 2015
COLLABORATIONS	IAS gravitational wave search pipeline CAMELS simulation collaboration Member of the Dark Energy Spectroscopic Instrument (DESI)	2022-current 2022-current 2019-2025
OUTREACH	• <i>Outreach talks:</i> Before the pandemic started, I used to give ~ 5 talks each year to high schools students in my hometown in India about the current cutting-edge research in science and ways of pursuing research as a career option. Here is an example • <i>Academic Mentorship:</i> - Tutored academically weak students at IIT Bombay in complex analysis and differential equations. Mentored two students in the physics department and helped them in clearing their backlogs. - Mentoring a JHU astrophysics graduate student as a part of PhA mentoring program at JHU. • <i>Astronomy Club:</i> Gave talks on future of astronomy at IIT Bombay to a general audience. I also headed a project in collaboration with the club to build a Solar Radio Telescope from scratch. • Completed science communication writing workshops at the NYU journalism institute and published a review on an upcoming popular science book [link].	
REFERENCES	<i>Prof. Matias Zaldarriaga</i> <i>Prof. Roman Scoccimarro</i> (PhD advisor) <i>Prof. Glennys Farrar</i> <i>Prof. Emanuele Berti</i> <i>Prof. David Spergel</i> <i>Prof. Shirley Ho</i> <i>Prof. Colin Hill</i>	matiasz@ias.edu rs123@nyu.edu gf25@nyu.edu berti@jhu.edu dspergel@flatironinstitute.org shirleyho@flatironinstitute.org jch2200@columbia.edu
RECENT CONTRIBUTED TALKS	GWPAW (Gravitational wave physics and astronomy workshop), GA GWPAW conference, University of Birmingham Gravitational wave group meeting, University of Cambridge, UK Cosmology journal club, University of Cambridge, UK April Meeting of the American Physical Society (APS), Sacramento, CA	December 2025 May 2024 May 2024 May 2024 April 2024

Astrophysics talk, JHU	January 2024
GRITTS seminar, MIT	December 2023
Tea talk, Caltech	November 2023
BCCP journal club, UC Berkeley	November 2023
Theory group seminar, Northwestern University	October 2023
APS April Meeting, Minneapolis, MN	April 2023
L2G2 meeting, Columbia University	March 2023
APS April Meeting, New York, NY	April 2022
Princeton/IAS cosmology meeting, Princeton University	October 2021
Brown bag talk, NYU	March 2021
Particle and Astrophysics meeting, CCA	December 2020
Cosmology group meeting, University of Chicago	December 2020
Astrophysics seminar, University of Pennsylvania	December 2020
Cosmology seminar, Caltech/JPL	November 2020
Cosmology group meeting, CITA	November 2020
Cosmology seminar, Perimeter	November 2020
Dvorkin group meeting, Harvard	November 2020
Eisenstein group meeting, Harvard	October 2020
Hernquist group meeting, Harvard	October 2020
Lunch talk, MIT	October 2020
Cosmology at home conference [video]	August 2020
Euclid survey ML meeting, Zoom	July 2020
BCCP workshop: Spectroscopic surveys, UC Berkeley, CA	January 2020
APS April Meeting, Denver, CO	April 2019
APS April Meeting, Columbus, OH	April 2018